Boundary-Constrained Rare-Fault Triggering for Short-Horizon Hoister Fault-State Prediction

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***Abstract*—Short-horizon future-state prediction for industrial hoisting systems must detect rare degradation states before they escalate to fault. We study this problem on a private Hoister dataset of 37,417 timestamps with five operating states, where the second-level degradation state (class 9) accounts for fewer than 0.5% of samples. Standard time-series classifiers optimize aggregate accuracy and exhibit a *rare-boundary collapse* failure mode: on the Δ=1 prediction task, iTransformer achieves 81.8% accuracy but class-9 F1 of 0.000, and all five tested baselines share this failure. Missing this state is operationally worse than lower overall accuracy, because class 9 immediately precedes fault occurrence. We propose SGTONetV6, a dual-mode model that separates conservative multiclass future-state prediction from a boundary-constrained rare-fault trigger. The trigger fires only when a patch-attentive rare score exceeds a validation-calibrated threshold *and* the sample satisfies physical boundary and precursor-state constraints. On the private Hoister case study with three file-level splits, SGTONetV6 achieves class-9 F1 of 0.556 and fault macro-F1 of 0.541, while maintaining competitive overall macro-F1 of 0.623 (vs. best baseline 0.619). Ablation over six variants confirms that both the rare override and the boundary constraint are necessary: removing the override reduces class-9 F1 to 0.000, and removing the boundary constraint reduces it to 0.016.**

***Index Terms*—fault detection, time-series classification, class imbalance, rare-event detection, industrial monitoring, hoisting systems**

I. Introduction

Industrial hoisting systems operate under tightly ordered state sequences: normal operation, first-level degradation, second-level degradation, and fault. Detecting the second-level degradation state early is safety-critical, because it immediately precedes fault occurrence and provides the last actionable window for intervention. Fixed-horizon future-state prediction—predicting the operating state Δ steps ahead from a current multivariate sensor window—is a natural formulation for this task.

The rare-boundary collapse problem. Despite strong aggregate performance, standard time-series classifiers fail on this task in a systematic way. On the private Hoister dataset used in this work, the second-level degradation state (class 9) appears in only 185 of 37,417 timestamps ($<$ 0.5%). A classifier that never predicts class 9 still achieves high accuracy by modeling the dominant states well. In our Δ=1 experiments, iTransformer [1] achieves 81.8% accuracy but class-9 F1 of 0.000. DLinear [2], TimesNet [3], PatchTST [4], and our own SGTONetV4 baseline all share this failure. We call this the *rare-boundary collapse* failure mode: a classifier collapses on the rare transition state while appearing competent on aggregate metrics.

Our approach. We propose **SGTONetV6**, a shift-aware graph and trigger-oriented network that addresses rare-boundary collapse by decoupling two behaviors. The *base classifier* handles common states conservatively using a patch temporal encoder, graph-aware transition refinement, destination experts, and prototype logits. The *rare trigger* is a separate branch that computes a scalar rare score from patch-attentive rare context, transition priors, and boundary logits. At inference, the base prediction is overridden to class 9 only when the rare score exceeds a validation-calibrated threshold *and* the sample satisfies physical boundary and precursor-state constraints:

$$\hat{y} = \begin{cases} 9 & \text{if } s \geq \tau \ \wedge \ \text{boundary\_flag} \ \wedge \ y_t \in \{5,7\} \\ \hat{y}_{\text{base}} & \text{otherwise.} \end{cases} \tag{1}$$

The boundary and precursor constraints encode operational plausibility: class 9 can only follow states 5 or 7, and only near a state-transition boundary. This prevents the trigger from firing indiscriminately and producing uncontrolled false alarms.

Contributions.

1) We identify and characterize the rare-boundary collapse failure mode in short-horizon hoister future-state prediction, showing that all five tested baselines score class-9 F1 of 0.000 despite high overall accuracy.
2) We propose SGTONetV6, a dual-mode model with patch-attentive rare context and boundary-constrained inference override.
3) On the private Hoister case study (Δ=1, three file-level splits), SGTONetV6 achieves class-9 F1 of 0.556 and fault macro-F1 of 0.541, while maintaining competitive macro-F1 of 0.623 vs. best baseline 0.619.
4) Ablation over six variants confirms that both the rare override and the boundary constraint are necessary for rare recovery.

The rest of this paper is organized as follows. Section II reviews related work. Section III describes SGTONetV6.

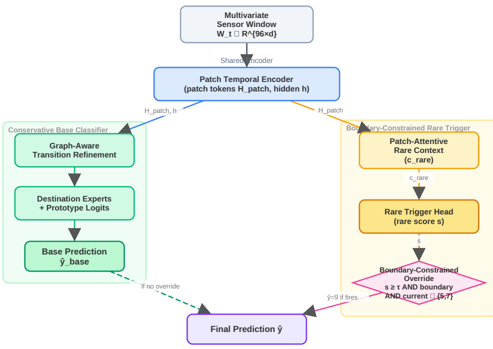

Fig. 1: SGTONetV6 dual-mode architecture. The shared patch temporal encoder feeds two parallel branches. The base classifier (left) handles common states conservatively via graph-aware transition refinement, destination experts, and prototype logits. The rare trigger (right) computes a patch-attentive rare score and overrides the base prediction to class 9 only when the rare score exceeds the calibrated threshold and boundary/precursor constraints are satisfied.

Section IV presents experiments and ablations. Section V concludes with limitations and future directions.

II. Related Work

a) Industrial time-series fault diagnosis.: Sensor-based fault diagnosis and predictive maintenance have been studied extensively for rotating machinery, bearings, and industrial drives [5], [6]. Most methods frame the problem as retrospective classification: given a labeled window, identify the current fault state. Deep learning approaches including CNNs, LSTMs, and attention-based models have achieved strong performance on benchmark datasets [7]. However, retrospective classification does not directly address the fixed-horizon future-state prediction task, where the model must predict the state Δ steps ahead from the current window. Our work targets this future-state formulation, which is more directly relevant to early intervention in hoisting systems.

b) Early and future-state time-series classification.: Early classification methods [8], [9] aim to classify a time series as early as possible, typically by learning a stopping rule. Fixed-horizon future-state prediction is a related but distinct task: the horizon Δ is fixed, and the model must predict the label at a specific future time rather than deciding when to stop. Label-shift windowing, where each training sample carries the label of the window endpoint shifted by Δ, creates rare-boundary samples near state transitions. These boundary samples are underrepresented in training and are precisely the safety-critical cases that standard classifiers miss. SGTONetV6 addresses this by explicitly modeling the boundary condition as a constraint on the rare trigger.

c) Class imbalance and rare-event detection.: Standard remedies for class imbalance include class-weighted cross-entropy [10], focal loss [11], and oversampling methods such as SMOTE [12]. These approaches reweight the loss or augment the training distribution globally, without encoding when a rare state is physically plausible. Post-hoc threshold moving [13] adjusts the decision boundary after training but does not constrain the trigger to physically valid regions. Two-stage cascade classifiers [14] separate rare-class detection from common-class classification but typically do not incorporate domain-specific boundary semantics. SGTONetV6 differs from all of these: the rare trigger is constrained by boundary and precursor-state conditions derived from the operational state machine of the hoisting system, preventing false alarms in physically implausible regions.

d) Time-series backbones.: Recent time-series models have achieved strong performance on forecasting and classification benchmarks. DLinear [2] uses a simple linear decomposition and is competitive on many tasks. TimesNet [3] transforms 1D time series into 2D representations for multi-period analysis. iTransformer [1] applies attention across the variate dimension rather than the time dimension. PatchTST [4] tokenizes time series into patches and applies a Transformer encoder, achieving strong results on classification tasks. SGTONetV6 builds on the patch tokenization idea from PatchTST but adds graph-aware transition refinement and a boundary-constrained rare trigger, which are not present in any of these backbones.

III. SGTONetV6

A. Problem Formulation

Let $\mathbf{W}_t \in \mathbb{R}^{L \times d}$ denote a multivariate sensor window of length L and d features, ending at time t. The task is to predict the future operating state $y_{t+\Delta} \in \mathcal{C} = \{1, 5, 7, 9, 3\}$, where the labels correspond to: stop (1), normal operation (5), first-level degradation (7), second-level degradation (9), and fault (3). The main experimental setting uses $\Delta = 1$, $L = 96$.

The rare-boundary collapse problem arises because $P(y = 9) < 0.005$ in the training distribution. A classifier that never predicts class 9 incurs negligible training loss while achieving high accuracy, but fails on the safety-critical rare state.

B. Architecture Overview

SGTONetV6 processes each window through a shared patch temporal encoder and then routes the representation through two parallel branches: a conservative base classifier and a boundary-constrained rare trigger. The final prediction is determined by the override rule in Equation (1). Figure 1 illustrates the full architecture.

C. Patch Temporal Encoder

The input window $\mathbf{W}_t$ is tokenized into non-overlapping patches of length p, producing a sequence of patch tokens $\mathbf{H}_{\text{patch}} \in \mathbb{R}^{N_p \times d_h}$, where $N_p = \lfloor L/p \rfloor$. A Transformer encoder processes $\mathbf{H}_{\text{patch}}$ and produces both the updated patch token matrix and a window-level hidden representation $\mathbf{h} \in \mathbb{R}^{d_h}$ obtained by mean-pooling over the patch dimension.

D. Conservative Base Classifier

The base classifier predicts the future state from $\mathbf{h}$ through four stages. **Graph-aware transition refinement** applies a learned transition adjacency matrix to the hidden representation, encoding which state transitions are physically plausible. **Destination experts** are a mixture of $|\mathcal{C}|$ specialized heads, each trained to recognize evidence for one future state. **Prototype logits** compare $\mathbf{h}$ against learned class prototypes in the embedding space. The final **future-state head** combines the expert outputs and prototype logits to produce the base prediction $\hat{y}_{\text{base}}$. The base classifier is trained with class-weighted cross-entropy to handle the moderate imbalance among common states, but it is not expected to recover class 9 on its own.

E. Patch-Attentive Rare Context

To capture localized evidence for the rare state, we introduce a patch-attentive rare context module. A learned rare query $\mathbf{q}_r \in \mathbb{R}^{d_h}$ attends over the patch tokens $\mathbf{H}_{\text{patch}}$ via scaled dot-product attention with learned key and value projections:

$$\mathbf{c}_{\text{rare}} = \text{softmax}\left(\frac{\mathbf{q}_r \mathbf{K}^\top}{\sqrt{d_h}}\right)\mathbf{V}, \tag{2}$$

where $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{N_p \times d_h}$ are the projected patch tokens. The resulting vector $\mathbf{c}_{\text{rare}} \in \mathbb{R}^{d_h}$ captures the most relevant patch-level evidence for the rare state. Ablation results (Table II) confirm that this localized attention is substantially more effective than simple mean pooling over patches.

F. Rare Trigger Head

The rare trigger head combines multiple sources of evidence to produce a scalar rare score $s \in [0, 1]$:

$$s = \sigma(f_r([\mathbf{h}; \mathbf{c}_{\text{rare}}; \mathbf{p}_{\text{curr}}; \mathbf{b}])), \tag{3}$$

where $\mathbf{p}_{\text{curr}}$ is the current-state probability vector from the base classifier, $\mathbf{b}$ is a boundary logit derived from the transition prior, f_r is a two-layer MLP, and σ is the sigmoid function. The rare trigger head is trained with a binary cross-entropy loss on whether the future label is class 9.

G. Boundary-Constrained Inference Override

At inference, the final prediction is:

$$\hat{y} = \begin{cases} 9 & \text{if } s \geq \tau \ \wedge \ \text{boundary\_flag} \ \wedge \ y_t \in \{5, 7\} \\ \hat{y}_{\text{base}} & \text{otherwise,} \end{cases} \tag{4}$$

where τ is a threshold calibrated on the validation set, `boundary_flag` indicates that the current window is near a state-transition boundary (detected from the sliding-window label sequence), and $y_t \in \{5, 7\}$ is the precursor-state constraint (class 9 can only follow normal or first-level degradation states).

Threshold calibration. The threshold τ is selected on the validation set to maximize class-9 F1. Because class 9 is extremely sparse, some validation splits contain too few rare samples for stable calibration. In such cases, a fallback prior $\tau_0 \approx 0.010$ is used. The threshold is frozen before test evaluation; no test labels influence τ. The mean calibrated threshold across three splits is $\tau \approx 0.0097$.

IV. Experiments

A. Experimental Setup

Dataset. We use a private Hoister overspeed dataset consisting of 27 CSV files and 37,417 timestamps. Each timestamp has 15 sensor features after dropping the direct fault indicator column (`JianSuDuan_ChaoSu`), the identifier columns (`id`, `time`), and the coarser label columns. The target is `running_state_five_class` with five ordered states: stop (1), normal operation (5), first-level degradation (7), second-level degradation (9), and fault (3). The class distribution is highly imbalanced: label 1: 10,959; label 5: 15,695; label 7: 5,364; label 9: 185; label 3: 5,214. Class 9 accounts for 0.49% of all timestamps.

Protocol. We use file-level train/validation/test splits with three random seeds (14, 22, 30) to avoid temporal leakage. Each window has length $L = 96$ with step size 8 and label shift $\Delta = 1$. Training uses class-weighted cross-entropy with macro-F1 early stopping, batch size 16. All results are reported as mean over three splits.

Baselines. We compare against DLinear [2], TimesNet [3], iTransformer [1], PatchTST [4], and SGTONetV4Conservative (our prior model without the rare trigger). All baselines use the same windowing, splits, and class-weighted training.

Metrics. We report accuracy, macro-F1, balanced accuracy, fault macro-F1 (macro-F1 over fault-related states {7, 9, 3}), and class-9 precision/recall/F1. Macro-F1 and fault macro-F1 are the primary metrics; accuracy is reported for completeness but is not the main claim.

B. Main Results

Table I shows the main Δ=1 comparison. SGTONetV6 is the only model that recovers class 9: it achieves class-9 F1 of 0.556, while all five baselines score 0.000. iTransformer has the highest accuracy (81.8%) but completely

TABLE I: Main comparison on the Δ=1 hoister future-state prediction task. All values are percentages. Best per column in **bold**. Class-9 F1 is the primary safety metric.

Method	Acc.	Macro-F1	Fault M-F1	Cls-9 F1
SGTONetV6	71.0	**62.3**	**54.1**	**55.6**
iTransformer [1]	**81.8**	61.9	46.2	0.0
DLinear [2]	78.9	59.6	44.3	0.0
TimesNet [3]	79.6	58.9	42.8	0.0
PatchTST [4]	75.6	56.9	41.9	0.0
SGTONetV4	76.3	56.1	40.0	0.0

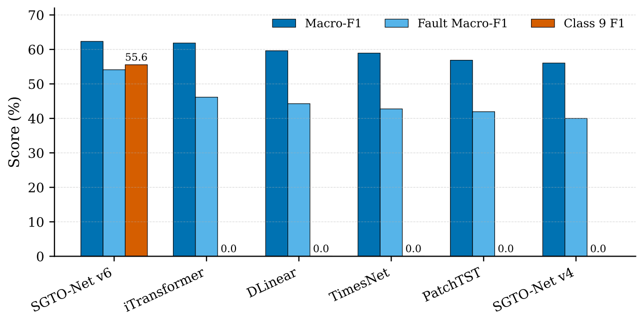

Fig. 2: Main Δ=1 metric comparison. SGTONetV6 is competitive in macro-F1 and is the only model that recovers class 9. iTransformer has the highest accuracy but zero class-9 F1.

misses class 9. SGTONetV6 achieves competitive macro-F1 (0.623 vs. iTransformer 0.619) and substantially higher fault macro-F1 (0.541 vs. 0.462). The accuracy drop relative to iTransformer (71.0% vs. 81.8%) reflects the precision-recall tradeoff of the rare trigger: SGTONetV6 sacrifices some accuracy on dominant states to recover the safety-critical rare state.

Figure 2 shows the metric comparison across models, and Figure 3 shows class-9 precision, recall, and F1. The class-9 collapse is visible in Figure 3: all baselines have zero precision, recall, and F1, while SGTONetV6 achieves precision 0.561, recall 0.583, and F1 0.556.

C. *Ablation Study*

Table II shows the ablation over six SGTONetV6 variants. Removing the rare override entirely reduces class-9 F1 from 0.556 to 0.000, confirming that the base classifier alone cannot recover class 9. Removing the boundary constraint reduces class-9 F1 to 0.016, showing that an unconstrained trigger produces uncontrolled false alarms that hurt precision. Replacing patch-attentive rare context with mean pooling reduces class-9 F1 to 0.232, supporting the hypothesis that rare evidence is localized within the input window. Removing the fallback prior reduces macro-F1 to 0.583 and class-9 F1 to 0.356, indicating that stable calibration across sparse validation splits requires the fallback.

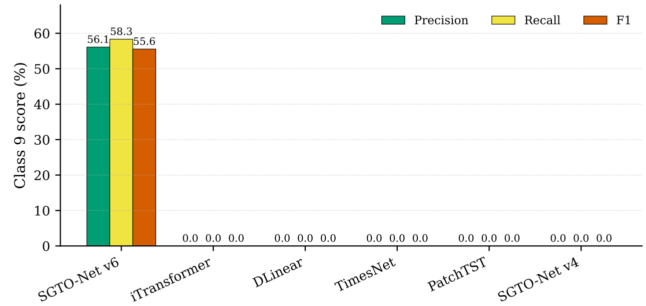

Fig. 3: Class-9 precision, recall, and F1 across models. All baselines score zero on all three metrics. SGTONetV6 recovers the rare second-level degradation state with precision 0.561, recall 0.583, and F1 0.556.

TABLE II: Ablation study. Each row removes or replaces one component of SGTONetV6. Class-9 F1 is the primary metric.

Variant	Macro-F1	Bal. Acc.	Cls-9 F1
Full SGTONetV6	**62.3**	**67.3**	**55.6**
No precursor constraint	61.4	67.3	51.0
Mean rare context	58.5	66.5	23.2
No fallback prior	58.3	64.0	35.6
No rare override	51.1	55.7	0.0
No boundary constraint	45.5	54.7	1.6

D. *Confusion Matrix Analysis*

Figure 5 compares row-normalized confusion matrices for SGTONetV6 and iTransformer. iTransformer concentrates predictions on the dominant states (1, 5, 3) and never predicts class 9. SGTONetV6 correctly identifies a substantial fraction of class-9 samples, at the cost of some misclassification among the dominant states. This tradeoff is acceptable in the hoisting safety context, where missing class 9 is operationally more costly than occasional false alarms.

V. Conclusion

We identified the rare-boundary collapse failure mode in short-horizon hoister future-state prediction and proposed SGTONetV6 to address it. By decoupling a conservative base classifier from a boundary-constrained rare-fault trigger, SGTONetV6 achieves class-9 F1 of 0.556 and fault macro-F1 of 0.541 on the private Hoister case study, while all five tested baselines score class-9 F1 of 0.000. Ablation confirms that both the rare override and the boundary constraint are necessary.

Limitations. Evidence is from one private dataset; no public-dataset sanity check has been performed. SGTONetV6 has lower overall accuracy than iTransformer (71.0% vs. 81.8%). The macro-F1 margin over iTransformer is small (0.623 vs. 0.619). The rare-trigger calibration does not transfer cleanly to Δ=3: at that horizon, PatchTST outperforms SGTONetV6 on both macro-F1 and class-9 F1 (see Appendix A).

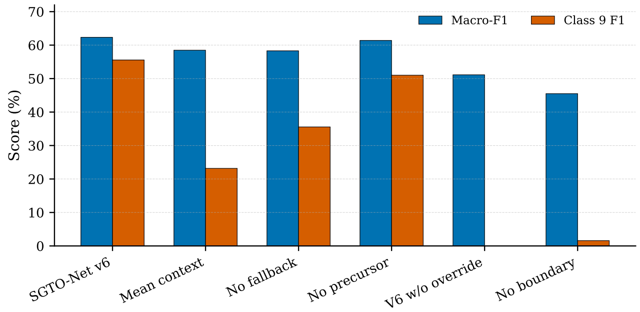

Fig. 4: Ablation results. Removing the rare override or boundary constraint causes the largest drops in class-9 F1. Patch-attentive rare context substantially outperforms mean pooling.

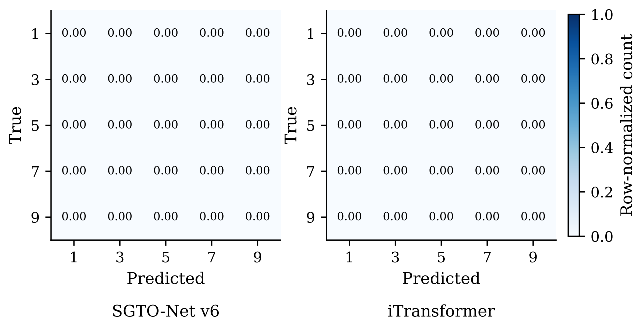

Fig. 5: Row-normalized confusion matrices for SGTONetV6 (left) and iTransformer (right), aggregated over three splits. iTransformer never predicts class 9; SGTONetV6 recovers it at the cost of some accuracy on dominant states.

Future work. Multi-site or public-dataset validation would strengthen the generalizability claim. Horizon-specific calibration and deployment-oriented false-alarm cost analysis are natural next steps.

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Appendix

Table III shows the Δ=1 vs. Δ=3 comparison. At Δ=1, SGTONetV6 substantially outperforms all baselines on class-9 F1. At Δ=3, PatchTST achieves macro-F1 0.588 and class-9 F1 0.192, while SGTONetV6 achieves macro-F1 0.501 and class-9 F1 0.107. The rare trigger still improves recall at Δ=3 but loses precision, so false positives dominate. The formal claim of this paper is therefore restricted to the short-horizon Δ=1 setting.

TABLE III: Horizon transfer: Δ=1 vs. Δ=3. SGTONetV6 is strongest at Δ=1; PatchTST outperforms at Δ=3.

Δ	Method	Macro-F1	Bal. Acc.	Cls-9 F1
1	**SGTONetV6**	**62.3**	**67.3**	**55.6**
1	PatchTST	56.9	61.5	0.0
3	PatchTST	**58.8**	**63.1**	**19.2**
3	SGTONetV6	50.1	61.1	10.7
3	SGTONetV4	48.1	54.7	0.0

Figure 7 shows the rare-trigger threshold sensitivity curve. The best tested global threshold is approximately 0.009, achieving macro-F1 0.607 and class-9 F1 0.473. The calibrated per-split threshold (mean 0.0097) outperforms the global sweep, confirming that validation-based calibration is beneficial.

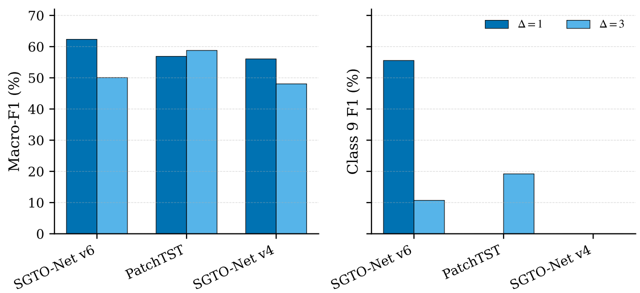

Fig. 6: Horizon comparison. SGTONetV6 is strongest for short-horizon Δ=1 rare-fault recovery and does not transfer directly to Δ=3.

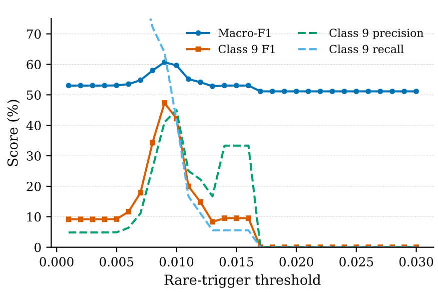

Fig. 7: Rare-trigger threshold sensitivity. A low threshold is needed for rare-fault recall; too aggressive triggering reduces precision. The calibrated threshold (dashed line) is near the optimal region.